1.1.3 Conception Phase

Sentiment analysis from news articles - Project: Artificial Intelligence (DLBDSEAIS02)

1. Project Goal and Scope

The primary goal of this project is to develop a sentiment analysis application for the marketing department. The system will analyze public perception of the company by identifying the sentiment (positive, neutral, or negative) of recent news articles based on a given keyword (e.g., the company name). This tool will provide actionable insights into market sentiment, helping to guide marketing strategy and public relations efforts.

2. System Architecture and Process

The application is designed as a modular, web-based tool that provides real-time feedback to the user. The process flow is designed to be robust and transparent, from data acquisition to final analysis.

2.1. Process Diagram

The following architecture diagram is included as a JPG image. The chart source is stored in a separate file so it can be edited independently and exported again when needed.

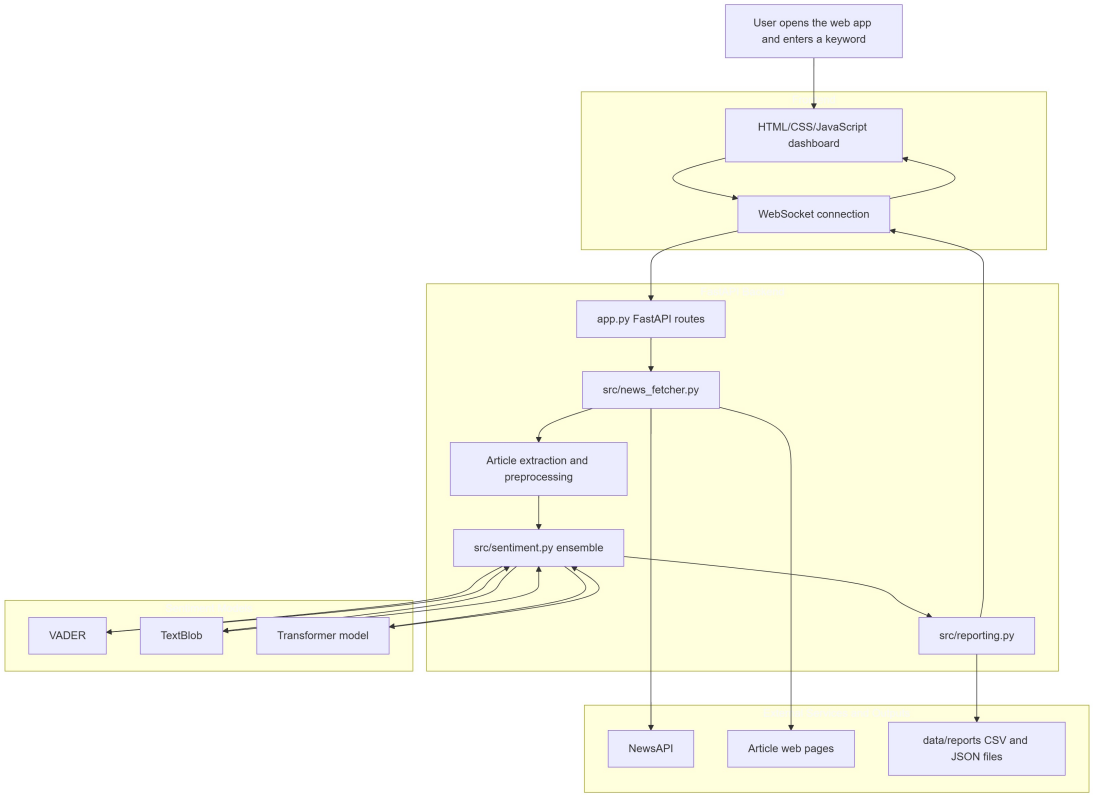

Figure 1: System Architecture and Data Flow

2.2. Process Steps

1. **User Input:** The user provides a keyword through a simple web interface.
2. **News Fetching:** The backend fetches a list of recent news articles related to the keyword from NewsAPI.
3. **Content Extraction:** For each article, the system navigates to the URL and extracts the full text content, stripping away HTML, ads, and other irrelevant elements.
4. **Sentiment Analysis:** The cleaned text is analyzed by an ensemble of three different sentiment analysis models.
5. **Reporting:** The results, including the sentiment from each model and a final consensus score, are saved to structured data files (CSV and JSON) for archival and further analysis.
6. **Feedback:** Live progress and the final results are streamed back to the user's web interface via WebSockets.

3. Classification and Validation Strategy

Given the subjective and nuanced nature of sentiment in news articles, relying on a single model can be unreliable. This project employs an ensemble-based approach for both classification and validation.

- **Classification:** The final sentiment label for an article is determined by a majority vote from the three models. This "consensus" approach mitigates the risk of a single model's error or bias dominating the result.
- **Validation:** The quality of the classification is validated internally by measuring the level of agreement between the models. A "confidence" score is calculated based on the proportion of models that agree on the consensus label (e.g., 1.0 if all three agree, 0.67 if two agree). A low confidence score indicates a contentious article that may require manual review, providing an inherent quality metric without needing a pre-labeled dataset. This method allows the marketing team to focus on high-confidence results and flag ambiguous ones.

This structure was chosen because it provides a more robust and trustworthy result than a single-model system. It delivers not just a classification, but also a measure of its reliability.

4. Frameworks and Tools

The technology stack was selected to be modern, efficient, and well-suited for NLP tasks and web-based delivery, using Python as the core language.

- **Backend Framework:** FastAPI was chosen for its high performance, asynchronous capabilities (essential for handling I/O-bound tasks like fetching articles), and native support for WebSockets, which enables a responsive user experience.
- **News Source:** NewsAPI provides a simple and effective way to source recent news articles from a wide variety of publications.
- **Content Extraction:** BeautifulSoup4 and Requests are used to download and parse the HTML of news articles, extracting the core text content.
- **Sentiment Models:**

  - **VADER** and **TextBlob**: These lexicon-based models are fast and provide a good baseline.
  - **Transformers (RoBERTa-based)**: A state-of-the-art model from Hugging Face (‘cardiffnlp/twitter-roberta-base-sentiment-latest‘) is used for its ability to understand context and nuance, providing a much more accurate analysis for complex sentences found in news.

- **Data Handling:** Pandas is used for creating and managing the final CSV reports, which are easily opened in Excel or other data analysis tools.

- **Executable Packaging:** PyInstaller is planned for packaging the application into a single standalone executable for easy distribution to the marketing department, removing any need for them to interact with Python or code.

This combination of tools provides a powerful and flexible foundation for building, validating, and deploying the sentiment analysis application effectively.